

Executive Summary

PASS RATE 28% target 90%	TOTAL TESTS 168 executed	PASSED 48 29% of total	FAILED 120 71% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 28% | Target: 90% | Gap: +62%

The Revenue Management System Kenya backend API testing has revealed significant stability and reliability concerns, with an overall pass rate of 28% against a target of 90%. The API automation suite shows a 27.5% pass rate (22/80) while backend smoke validation achieved 29.5% (26/88), indicating systemic issues across multiple endpoints and core functionalities. These results suggest that the backend services are experiencing widespread failures in response generation, data processing, or service availability, requiring immediate technical investigation to identify root causes and implement corrective measures to achieve production readiness.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	22	54	29%	FAIL
2	BE Smoke Validation	88	26	62	30%	FAIL
TOTAL		168	48	120	28%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#68	17 Jul	168	28%	120	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	48	28.6%	180ms	Within SLA
Requests Failed	120	71.4%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

QUALITY INDICATORS

Severity	Count	%	Indicator	Value	Target
Critical	6	5%	Endpoint Coverage	88%	85%
High	8	7%	Execution Rate	100%	100%
Medium	4	3%	Contract Validation	95%	90%
Low	2	2%	Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on failed endpoints to identify common failure patterns and error response codes	Immediate	P1
2	Review backend service logs and monitoring data to identify performance bottlenecks, timeout issues, or resource constraints	3 days	P1
3	Validate API contract compliance and ensure response schemas match expected specifications across all endpoints	5 days	P2
4	Verify database connections, third-party service integrations, and authentication/authorization mechanisms are functioning correctly	1 week	P2
5	Implement enhanced error handling and logging mechanisms to provide more detailed failure diagnostics	1 week	P2
6	Prioritize fixing high-impact endpoints that affect critical revenue management workflows before addressing edge cases	2 weeks	P3

KEY FINDINGS

1	Critical gap between current pass rate (28%) and target threshold (90%) indicates backend services are not meeting reliability standards
2	API automation failures (72.5% failure rate) suggest systematic issues with endpoint responses, data contracts, or service dependencies
3	Backend smoke validation failures (70.5% failure rate) indicate core functionality and basic operations are compromised
4	Consistent failure patterns across both test suites point to underlying infrastructure or configuration issues rather than isolated endpoint problems
5	High failure concentration suggests potential issues with service integration, database connectivity, or authentication mechanisms

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation		29%
BE Smoke Validation		30%